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How higher education affects corporate human capital investment: Based on Upper Echelons Theory

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ABSTRACT

This study uses listed companies from China's Shanghai and Shenzhen stock markets to examine how the level of higher education attained by corporate executives affects the company's human capital investment, based on Upper Echelons Theory. The findings suggest a positive correlation: higher education levels among executives lead to increased investment in human capital. Further analysis reveals that this effect is most pronounced in large-scale and non-state-owned enterprises, suggesting that the benefits of higher education on human capital investment are facilitated by the company's scale and institutional flexibility.

1. Introduction

Corporate human capital investment is a strategic investment in employees, including expenses in areas such as recruitment, training, compensation, and benefits. By actively recruiting and introducing high-quality talent, companies can build strong teams and enhance their overall business level. Training programs help continuously improve employees' professional skills and leadership, adapting to market changes and technological advancements. Reasonable compensation and benefits not only motivate employees but also retain talent and reduce turnover rates. These investments are not just for current business needs but also for future sustainable development. By creating a good working environment and offering development opportunities, companies can attract and retain high-performing employees, thereby enhancing overall productivity and innovation capability.

The higher education background of the executive team has a profound impact on corporate human capital investment. From the perspective of Upper Echelons Theory, this reflects the leadership's demand for intelligence, creativity, and strategic thinking. Executives with a higher education background usually possess a richer knowledge base and can flexibly use advanced management theories and practices in strategic decision-making, thus showing more comprehensive and strategic thinking in human capital investment. The disciplinary background of executives may lead to differences in human capital investment, such as those in engineering or technology fields focusing more on technical training and innovative investment. This helps guide the company in building a high-quality, comprehensively developed team, improving the organization's innovation and competitiveness.

The study reveals that executives' educational backgrounds significantly and positively impact corporate human capital investment. The heterogeneity analysis indicates that this positive effect is particularly strong in small-scale and non-state-owned enterprises.

The article's primary contributions include systematically validating the significant positive influence of executives' educational

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backgrounds on human capital investment within corporations. This finding provides important empirical evidence for the field of organizational management, emphasizing the close connection between the executive team's educational level and human capital investment, thus guiding companies to enhance talent quality and overall competitiveness. Second, further heterogeneity analysis reveals that the positive impact of executives' educational background on corporate human capital investment is more significant in small-scale and non-state-owned enterprises. This discovery has guiding significance for companies to optimize management strategies under different scales and natures, providing important references for various companies to formulate more precise human capital investment strategies. Third, the study's results offer insights for corporate management practice, suggesting that the consideration of executive educational background should be a key factor in human capital investment decisions. Companies should focus more on the educational background of their executive team when selecting and training, to facilitate more beneficial strategic decisions for talent cultivation and development. This has practical guidance value for improving corporate performance, strengthening innovation capabilities, and adapting to market changes.

2. Literature review

2.1. Corporate human capital investment

Most research on human capital investment focuses on macro-level aspects, primarily highlighting its economic consequences. There is a notable gap in micro-level studies, especially those examining factors influencing corporate human capital investment. At the macro level, international scholars generally agree that enhancing societal human capital positively impacts industrial structure, stable economic growth, and national strength. For example, Park (2006) analyzed human capital's effect on economic growth from a distribution perspective, finding that imbalances in human capital can negatively affect growth. Maitra (2016) studied Singapore's human investment, employment, and economic growth over thirty years, revealing that initial investments had a modest impact but became more significant over time, supporting stable economic growth. Sakamoto (2018) categorized industrialized countries and found that those with higher human capital investment exhibited greater labor productivity. Haini (2019) examined data from ten Southeast Asian countries and concluded that upgrading human capital positively influences economic growth, and that human capital and the internet can synergistically boost a country's economy.

2.2. Educational background of executives

Executives' educational levels determine their ability to understand complex situations, process information, and identify opportunities, which in turn fosters innovative thinking. Higher education levels enhance managers' abilities to recognize external environmental information, explore innovative opportunities, and embrace research and development projects. In contrast, lower education levels correlate with reduced sensitivity to environmental changes. Qian et al. (2013) found that executives with higher education levels manage companies more effectively, respond accurately to market changes, and are more willing to invest in research and development. Tihanyi (2000) noted that highly educated executives allocate resources more efficiently, leading to better resource utilization and improved corporate performance. Barker and Mueller (2002) demonstrated that executives with relevant professional backgrounds significantly promote continuous investment and effective use of research and development funds. Cho et al. (2017) suggested that executives' academic backgrounds can mitigate information asymmetry, reducing information risk and debt agency costs, thereby facilitating easier access to loans.

In summary, most existing research on human capital investment revolves around the economic consequences of improved human capital investment and lacks studies on the micro-level factors affecting human capital investment. This provides a research opportunity for this article. The higher education background of executives has a significant impact on human capital investment. According to the Upper Echelons Theory, executives who have received higher education typically have a broader knowledge base, leadership skills, and strategic insight. This makes them more inclined to implement effective human capital strategies, improving overall organizational performance through the development of employee skills and knowledge. Higher education enables executives to better understand complex business challenges, respond more flexibly to market changes, and formulate more innovative and forward-looking strategies in human resource management. Therefore, this article empirically tests how higher education affects human capital investment based on micro-enterprises, taking the level of higher education received by corporate executives as the starting point.

3. Data and variables

3.1. Data sources

This study focuses on non-financial and non-real estate companies listed on China's Shanghai and Shenzhen A-share capital markets from 2013 to 2022. The initial dataset was refined through the following steps: (1) excluding ST and PT companies; (2) removing companies with liabilities exceeding their assets; (3) excluding cases where the corporate income tax rate is greater than 1 or <0 to ensure data validity; (4) removing samples with missing key variables; and (5) applying Winsorization to all continuous variables at the 1 % to 99 % levels. Data on corporate human capital investment was sourced from the Wind database, information on executives' higher education backgrounds was obtained from the CNRDS database, and other financial data primarily came from the CSMAR database.

3.2. Variables

3.2.1. Higher education

Education plays a crucial role in shaping individuals. Those with higher education not only possess strong professional skills but also excel in various other areas. As a result, highly educated executives can make swift and accurate decisions in complex and dynamic market environments and tend to have robust stress resilience. The key variable in this study is the education level of executives. Based on an extensive review of relevant literature and considering data availability, this study categorizes the education levels of middle and senior executives into five tiers: secondary vocational and below, junior college, bachelor's, master's, and doctorate and above, assigning values from 1 to 5, respectively.

3.2.2. Corporate human capital investment

Human capital refers to the value derived from investments in people, which enhances human resources and influences future economic and consumption capabilities. It encompasses the totality of knowledge, skills, physical abilities, and health that hold economic value. In this study, we apply the concept of data dimension reduction through principal component analysis to streamline four indicators: the proportion of employees with a bachelor's degree or higher (Edu), the proportion of R&D personnel (Tech), the proportion of technical personnel (Support), and per capita income generation (Lnrev). The first principal component extracted from this analysis serves as the proxy variable for human capital (Human_cap).

3.2.3. Control variables

The control variables selected for this study include company size (Size), measured by the natural logarithm of total business revenue; asset-liability status (Lev), represented by the ratio of total liabilities to total assets; growth capability (Growth), assessed by the net profit growth rate, calculated as the current year's net profit minus the previous year's net profit, divided by the previous year's net profit; proportion of institutional investors (Insinvestorprop), measured by the ratio of shares held by institutional investors to the total number of company shares; earnings per share (PerIncome), calculated as the ratio of net profit to the number of ordinary shares; return on assets (Roa), measured by the ratio of net profit to total assets; equity checks and balances (OwnBalance), measured by the ratio of shares held by the second to tenth largest shareholders to the shares held by the largest shareholder; and fixed asset intensity (Tan_rate), represented by the ratio of the net value of fixed assets to total assets.

4. Empirical results

4.1. Descriptive statistics

Table 1 presents the descriptive statistics of the main variables in this study. The proxy variable for executives' higher education levels (Degree) ranges from a minimum of 1 to a maximum of 5, indicating significant variation in the educational attainment of executives within the sample. The proxy variable for corporate human capital investment (Human_cap) shows a minimum value of -2.0031, a maximum of 3.8846, a mean of -0.0073, and a median of 1.3026, suggesting considerable variability in human capital investment levels and an overall relatively low investment. The descriptive statistics for other variables align with findings from existing research and are not reiterated here.

4.2. Baseline results

To investigate how higher education affects corporate human capital investment, this paper constructs the following econometric model:

$$\text{Human\_cap}_{\text{it}} = \alpha + \beta_1 \text{Degree}_{it} + \gamma \text{ Controls}_{it} + \delta_I + \lambda_Y + \varepsilon_{it} \tag{1}$$

In this model, δ_I and λ_Y represent industry fixed effects and year fixed effects, respectively, while ε_{it} denotes the random disturbance term. The control variables, Controls_{it}, are included as previously described.

Table 1
Descriptive statistics.

VarName	Obs	Mean	SD	Min	Median	Max
Degree	11,050	3.6174	0.9322	1	4	5
Human_cap	11,050	-0.0073	0.9963	-2.0031	1.3026	3.8846
Size	11,050	21.9758	1.051	19.28	21.87	24.66
Lev	11,050	0.3182	0.1835	0.0341	0.2909	0.8704
Roa	11,050	0.0452	0.7271	-0.3113	0.1467	0.2629
Growth	11,050	0.3525	0.9382	-0.4361	0.1214	6.1553
PerIncome	11,050	0.6999	0.9208	-1.228	0.4697	5.606
OwnBalance	11,050	0.0778	0.09375	-0.6517	0.0649	0.9686
Tan_Rate	11,050	0.2613	0.1216	0.0067	0.2468	0.6414
Insinvestorprop	11,050	0.1924	0.1291	0.0155	0.1701	0.6094

The empirical test results of Model (1) are presented in Table 2. Column (1) shows the results with only year and industry fixed effects, while Column (2) includes a series of control variables. The proxy variable for executives' higher education levels and the proxy variable for corporate human capital investment are both significant at the 1 % level in both columns. This indicates that higher education levels among corporate executives are positively associated with increased human capital investment by the company. These findings suggest that higher education plays a beneficial role in promoting corporate investment in human capital.

4.3. Robustness test

4.3.1. Two-way fixed effects model

In addition to being influenced by common industry factors, a company's investment in human capital is also affected by unobservable individual characteristics such as corporate culture, atmosphere, and employee loyalty. Therefore, this paper further employs a two-way fixed effects model to re-examine the relationship between higher education and corporate human capital investment. The test results are shown in Table 3. It can be observed that the regression coefficients of the proxy variable for the higher education level of senior management and the proxy variable for corporate human capital investment are consistently significantly positive at the 5 % level. This indicates that the previously discussed conclusions are robust.

4.3.2. Excluding the impact of exogenous shocks

Considering the outbreak of the COVID-19 pandemic in China at the end of 2019, which severely impacted economic operations and caused social disruptions, this study excludes samples from after 2019 to avoid the interference of this public health emergency on the research conclusions. The model (1) was re-examined with the updated data, and the results are presented in Table 4. As shown in the results, the regression coefficients for the proxy variables of the level of higher education of corporate executives and the proxy variables for corporate human capital investment are significantly positive at the 5 % and 1 % levels, respectively, consistent with the previous research conclusions.

4.4. Cross-sectional analysis

4.4.1. Heterogeneity based on company size

The impact of higher education on corporate human capital investment involves the relationship between the educational level of the executive team and the size of the company. Under the framework of Upper Echelons Theory, corporate executives who have received a higher level of higher education may possess a richer knowledge and management skills, and thus be more aware of the importance of human capital investment for corporate development. However, it's worth delving into whether this conclusion has different effects due to the varying sizes of the sample companies. On one hand, large corporations face more complex organizational structures and management needs, and executives with higher levels of higher education can better meet these challenges, thereby leading the company to increase investment in human capital. On the other hand, small and medium-sized enterprises may focus more on flexibility and efficiency, and the educational level of executives may have a relatively small impact on human capital investment. Based on this, this article divides the samples into groups according to the annual industry median market value of the companies, and the group test results are shown in Table 5. It can be found that the impact of the higher education level received by corporate executives on corporate human capital investment is mainly reflected in large-scale companies. This indirectly indicates that in large-scale companies, ample resources and a sound institutional system help to play the scale effect of human capital investment. Therefore, in large-scale companies, the higher the level of higher education received by executives, the more inclined the company is to expand human capital investment.

4.4.2. Heterogeneity based on ownership type

This study, originating from the Upper Echelons Theory, delves into the relationship between the higher education received by corporate executives and human capital investment. Higher education, as a significant form of human capital, plays a crucial role in shaping the knowledge structure of a company, enhancing management levels, and driving innovation. The results indicate that the

Table 2
Baseline results.

	Human_cap	
	(1)	(2)
Degree	0.1326*	0.1518*
	(1.7482)	(1.8990)
Controlvariable	No	Yes
Year	Yes	Yes
Industry	Yes	Yes
N	11,050	11,050
R-Square	0.2431	0.2642

Note:***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively. *t*-values are displayed in parentheses.

Table 3
Two-way fixed effects model.

	Human_cap	
	(1)	(2)
Degree	0.1037**	0.1251**
	(2.2953)	(2.3801)
Controlvariable	No	Yes
Year	Yes	Yes
Firm	Yes	Yes
N	11,050	11,050
R-Square	0.2101	0.2302

Note:***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively. *t*-values are displayed in parentheses.

Table 4
Excluding the impact of exogenous shocks.

	Human_cap	
	(1)	(2)
Degree	0.1006**	0.1065***
	(2.0442)	(2.8990)
Controlvariable	No	Yes
Year	Yes	Yes
Industry	Yes	Yes
N	7823	7823
R-Square	0.2381	0.2942

Note: ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively. *t*-values are displayed in parentheses.

Table 5
Heterogeneity based on company size.

	Human_cap	
	Large-scale enterprises (1)	Small-scale enterprises (2)
Degree	0.1301*	0.1105
	(1.9421)	(1.5983)
Controlvariable	Yes	Yes
Year	Yes	Yes
Industry	Yes	Yes
N	5798	5252
R-Square	0.2315	0.2621

Note:***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively. *t*-values are displayed in parentheses.

higher the level of higher education received by executives, the more willing the company is to invest in human capital, reflecting the positive impact of knowledge and managerial capabilities on corporate strategy. However, the nature of ownership, as a core element of corporate governance structure, may have a differential impact on this relationship. In state-owned enterprises, the management may be constrained by more explicit organizational hierarchies and power structures, and the promoting effect of higher education on human capital investment might be regulated by relatively stable organizational mechanisms. Conversely, in non-state-owned enterprises, the emphasis on market competition and innovation capabilities may make the role of higher education in human capital investment more significant. Based on this, this paper groups the samples according to the nature of their ownership. The results of the group tests, as shown in Table 6, reveal that the positive impact of executives' higher education level on corporate human capital investment is mainly concentrated in non-state-owned enterprises. This indirectly indicates that in non-state-owned enterprises, a more flexible institutional system is more likely to allow higher education to enhance the investment in corporate human capital.

5. Conclusion

Human capital investment is of strategic significance to companies, as it not only enhances employee productivity and creativity but also helps build a robust corporate culture and employer brand, attracting and retaining top talent. This is also one of the keys for businesses to maintain competitiveness in today's fiercely competitive commercial environment. The findings of this study indicate

Table 6
Heterogeneity based on ownership type.

	Human_cap	
	Non-state-owned enterprises (1)	State-owned enterprises (2)
Degree	0.0801*	0.0615
	(1.6721)	(1.3083)
Controlvariable	Yes	Yes
Year	Yes	Yes
Industry	Yes	Yes
N	5328	5722
R-Square	0.1835	0.1641

Note:***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively. *t*-values are displayed in parentheses.

that the higher the level of higher education received by corporate executives, the greater the level of human capital investment by the company. This conclusion remains robust after a series of stability tests. Furthermore, the results of heterogeneity analysis suggest that the promoting effect of the higher education level of corporate executives on corporate human capital investment is mainly evident in large-scale and non-state-owned enterprises. This indirectly indicates that the positive impact of higher education on corporate human capital investment may be realized through the company's own scale effects and flexible institutional arrangements.

CRediT authorship contribution statement

Limeng Wu: Writing – review & editing, Supervision, Software, Resources, Formal analysis. **Jing Fang:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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